

Benjamin Lobos Lertpunyaroj

📞 765-532-8892 — ✉ blobosle@gmail.com — 🌐 blobosle.com — 💼 blobosle.com/linkedin — 🏠 github.com/Blobosle

Education

Purdue University

Bachelor of Science in Computer Science

Concentrations: Software Engineering, Systems Programming

Minor: Business Management

– College of Science Deans List & Semester Honors: 2023, 2024, 2025

– Relevant Coursework: *Data Structures and Algorithms, OOP, Discrete Mathematics, Linear Algebra, Software Engineering and Testing, Computer Architecture (x86, Arm64), C programming, Systems Programming, Operating Systems.*

West Lafayette, IN

May 2027

GPA: 3.85/4.0

Skills

Languages C, C++, C#.Net, Python, JavaScript (TS, React), HTML/CSS (TailwindCSS), Java, Swift, Lua, VimScript, R, Bash, AppleScript, Assembly (x86, Arm64)

Tools LaTeX, Vim/Neovim, Zsh, Apache Airflow, Linux, Debian, PostgreSQL, AArch64, Ubuntu, UNIX, Unity, Git, GitHub, CMake

Experience

Embedded Software Engineer @ ESAP

August 2025 - Present

- Engineered a C++ Hardware-in-the-Loop (HIL) simulation engine built using CMake on a Raspberry Pi, implementing Protobuf telemetry serialization to process simulated IMU, altimeter, and LiDAR data through a SPI connection for UAV controller testing.
- Architected a deterministic embedded data-exchange layer bridging the Raspberry Pi, ESP32-S3 (DUT), and Nucleo H563ZI using DMA-driven SPI and I2C protocols featuring CRC-validated framing and synchronized GPIO signaling.
- Led development on a modular Zephyr I2C target emulator for physical sensor spoofing, utilizing atomic primitives and snapshot caching.

C Programming Teaching Assistant @ Purdue University

November 2024 – Present

- Provided academic support to 750+ students (CS 24000), leading lab sessions, holding weekly office hours, and clarifying concepts related to the C programming language
- Developed homework assignments with custom built test-cases in C to reinforce data structure fundamentals, writing up handouts using LaTeX.
- Implemented multiple midterm/final practice exams into the class curriculum to enhance students' exam readiness and practical skills.

Software Engineering Intern @ Qservus

December 2022 – March 2023

- Shipped production-facing frontend features for Qservus Lite, a customer feedback and service-quality platform, using AngularJS.
- Enhanced deployment workflows for the international launch of Qservus's main web platform, improving release efficiency and readiness for broader adoption.
- Developed a native iOS survey application in Swift, expanding mobile product capabilities and streamlining customer feedback collection.

Projects

BoilerMate — TypeScript (React), PostgreSQL, Supabase, Prisma ORM, NestJS, AWS, AI

August 2025 – January 2026

Roommate Matching Solution

- Designed a roommate discovery website to streamline Purdue University's off-campus housing search, leveraging a NestJS/PostgreSQL backend to deliver a social media implementation powered by LLMs (Google AI Studio).
- Led coordination and onboarding of team members through Agile/SCRUM principles, shipping with over 5,000+ impressions and partnering with Purdue faculty stakeholders for expanding outreach.

Elite Edge — JavaScript (React), TailwindCSS, Python, PostgreSQL, Apache Airflow, Nginx

May 2025 - February 2026

College Basketball Weekly Insights & Analytics

- Architected a Dockerized backend to scrape weekly KenPom data (2013–2025), store it in PostgreSQL, and schedule a Python (scikit-learn) model running with Apache Airflow and a custom backend Nginx server for delivering NCAA advanced analytics.
- Constructed Machine Learning models (KNN, XGBoost) analyzing 12 years of data with 30+ parameters to predict tournament team selection and Final Four advancement.

Custom UNIX Shell — C++, C, UNIX, Flex, Bison

September – November 2025

Shell Command Line Tool for UNIX-based OS

- Built a Unix-based shell in C++ that tokenizes/parses user input (Flex/Bison) and executes commands via POSIX system calls, supporting arguments and standard I/O redirection.
- Engineered support for advanced features including interrupts, zombie process handling, builtin functions, environment variables, regex, and history, with an optimized memory allocator for running processes made in C.